

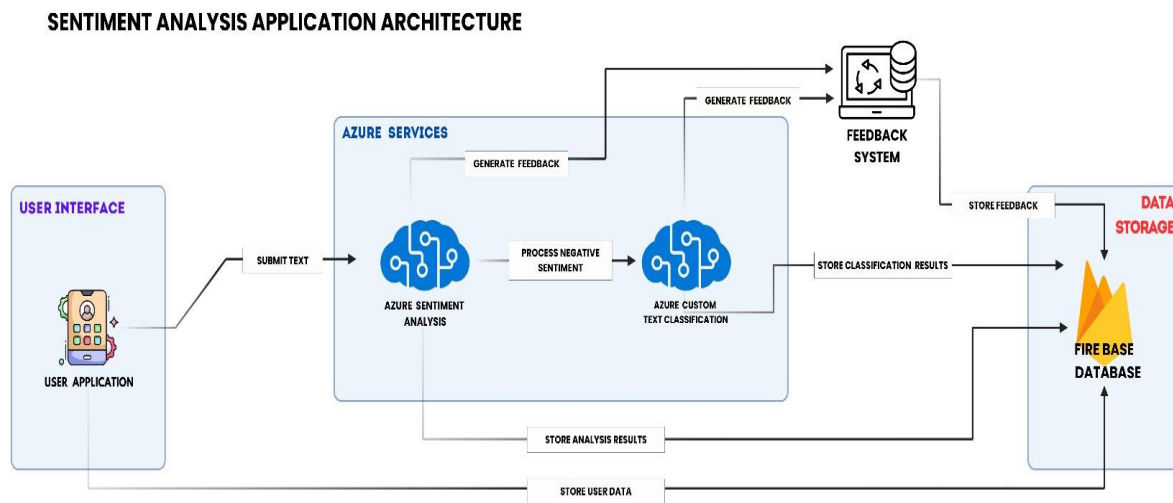
Supplementary Material

for

A Cloud-Based Two-Layer Text Classification Framework for Mental Health Screening with Sarcasm and Emoji-Aware Sentiment Analysis

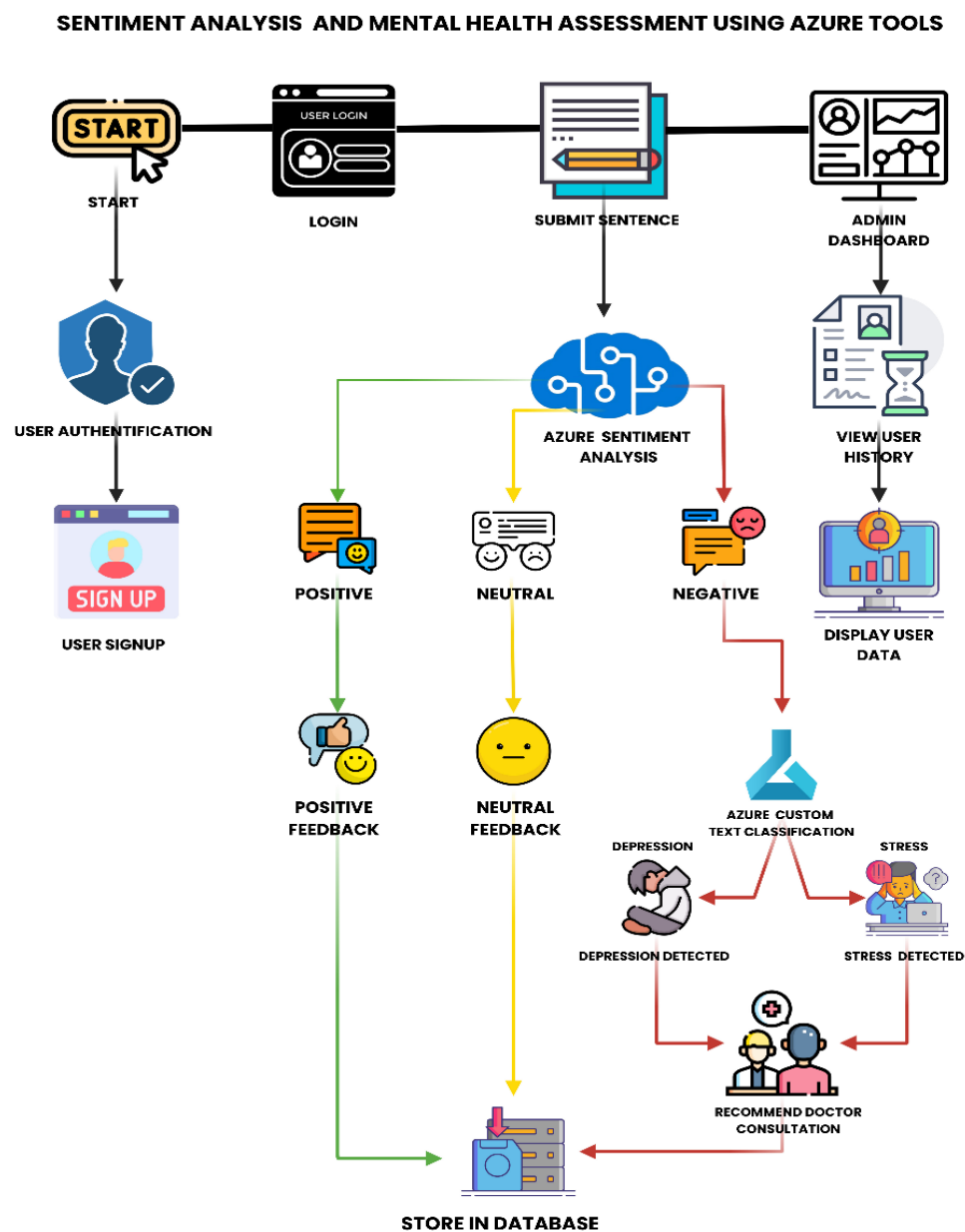
FIGURES:

FIGURE. 01: Overview architecture diagram of the proposed framework



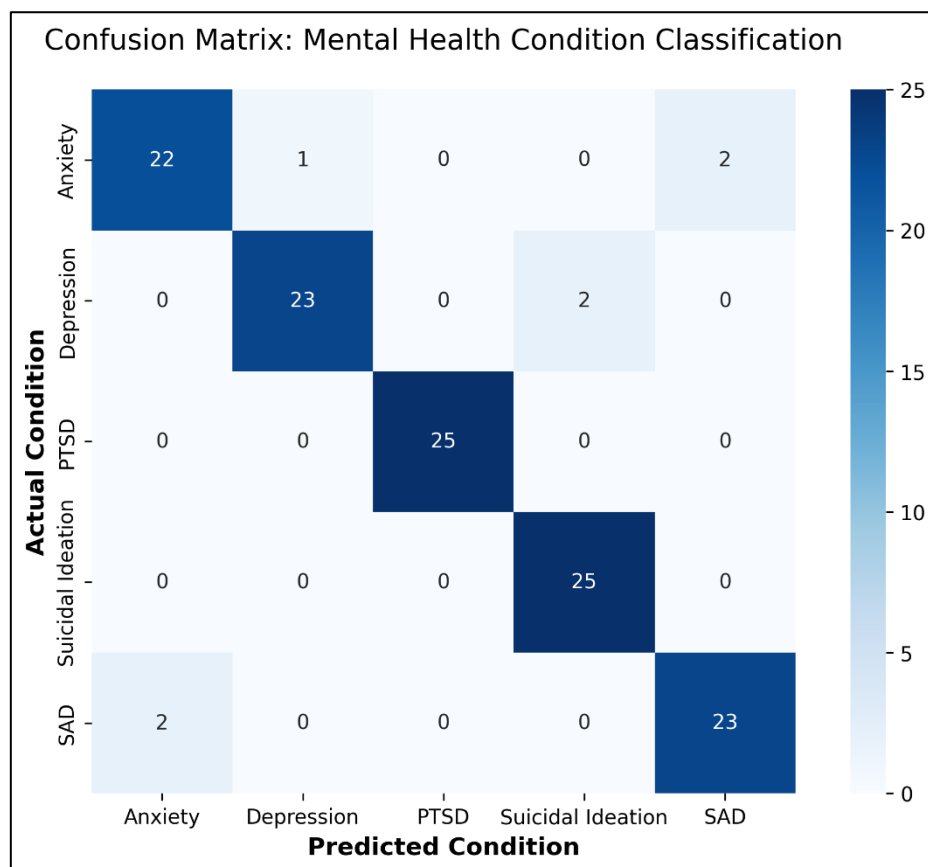
Description: The image represents the overall architecture of the proposed framework, illustrating the complete workflow from user authentication and text submission through Azure-based sentiment analysis, mental health-related text classification, feedback generation, and secure data storage. The figure also highlights the interaction between the Android application, Azure cognitive services, and Firebase infrastructure that collectively enable real-time processing and monitoring.

FIGURE. 02: Sentiment Analysis and Mental Health-Related Text Classification Using Azure Services



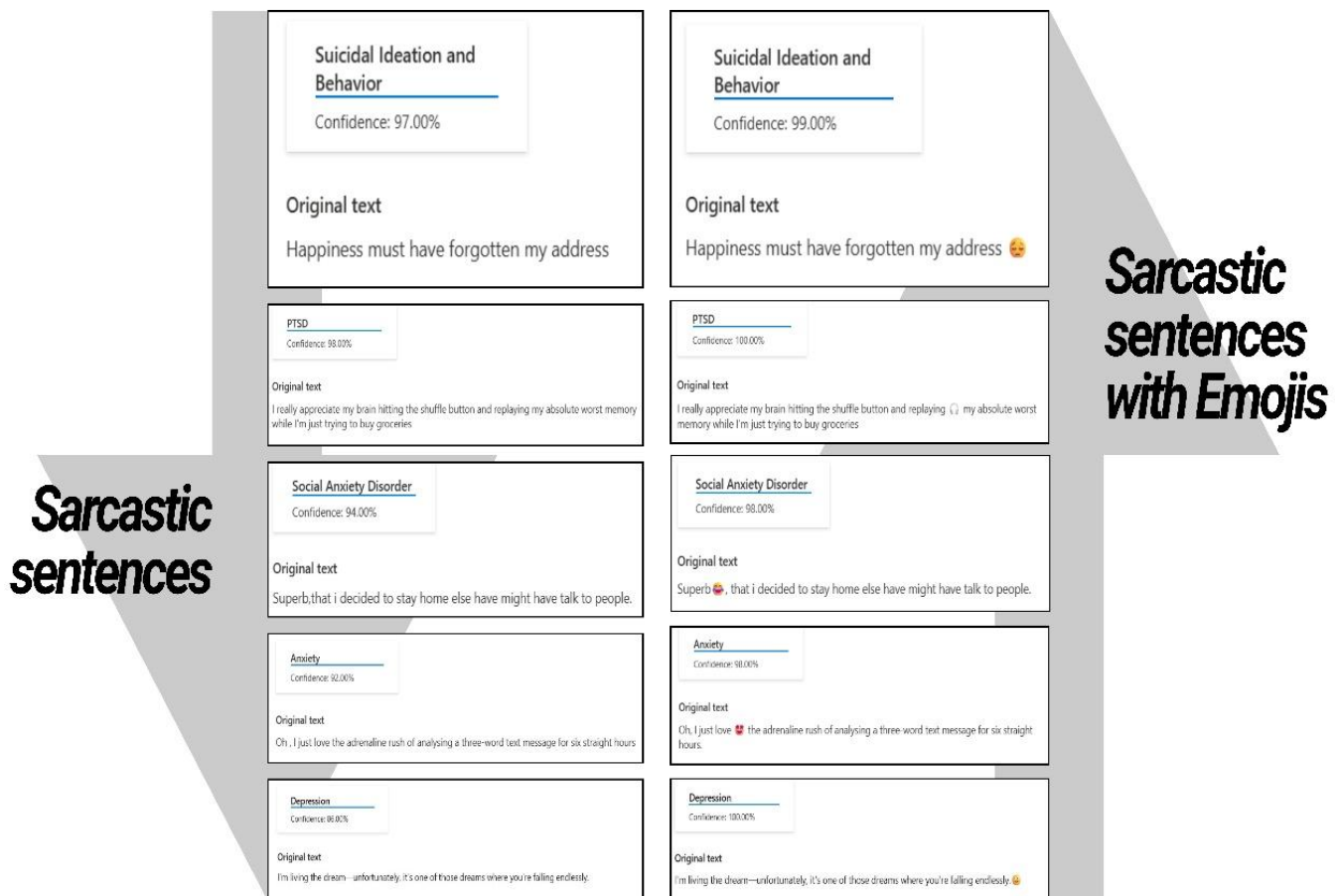
Description: The image illustrates the complete workflow of the proposed framework, beginning with user authentication and ending with secure storage and administrative monitoring. The process starts when a user registers and logs into the Android application. Following authentication, the user submits free-form textual input, which is transmitted securely to Azure cloud services for analysis.

FIGURE. 03: Confusion matrix obtained using the clinically reviewed validation dataset (125 samples)



Description: The matrix displays the distribution of actual versus predicted mental health conditions across 125 synthetic text samples (25 per category). The dark diagonal signifies high true-positive rates across all classes, with the model achieving perfect classification (25/25) for Post-Traumatic Stress Disorder (PTSD) and Suicidal Ideation. Off-diagonal values illustrate minor misclassifications occurring primarily between conceptually related conditions, such as the semantic overlap between Anxiety and Social Anxiety Disorder (SAD), as well as between Depression and Suicidal Ideation.

FIGURE. 04: Qualitative evaluation of sarcastic and informal language in mental health classification, and the influence of emojis on model confidence.



Description: The left column shows original sarcastic sentences, while the right column shows the same sentences augmented with emojis. The framework successfully identified underlying mental health categories (Suicidal Ideation and Behaviour, PTSD, Social Anxiety Disorder, Anxiety, and Depression) despite surface-level sarcasm and indirect expressions. Adding contextually relevant emojis consistently increased classification confidence while preserving the predicted category.

Post Traumatic Stress Disorder (PTSD) :

PTSD

Confidence: 98.00%

Original text

I really appreciate my brain hitting the shuffle button and replaying my absolute worst memory while I'm just trying to buy groceries

PTSD

Confidence: 100.00%

Original text

I really appreciate my brain hitting the shuffle button and replaying 🎧 my absolute worst memory while I'm just trying to buy groceries

Anxiety:

Anxiety

Confidence: 92.00%

Original text

Oh , I just love the adrenaline rush of analysing a three-word text message for six straight hours

Anxiety

Confidence: 98.00%

Original text

Oh, I just love 🥳 the adrenaline rush of analysing a three-word text message for six straight hours.

Suicidal Ideation and Behaviour :

Suicidal Ideation and Behavior

Confidence: 97.00%

Original text

Happiness must have forgotten my address

Suicidal Ideation and Behavior

Confidence: 99.00%

Original text

Happiness must have forgotten my address 🙄

Social Anxiety Disorder (SAD) :

Social Anxiety Disorder

Confidence: 94.00%

Original text

Superb,that i decided to stay home else have might have talk to people.

Social Anxiety Disorder

Confidence: 98.00%

Original text

Superb😄, that i decided to stay home else have might have talk to people.

Post Traumatic Stress Disorder (PTSD) :

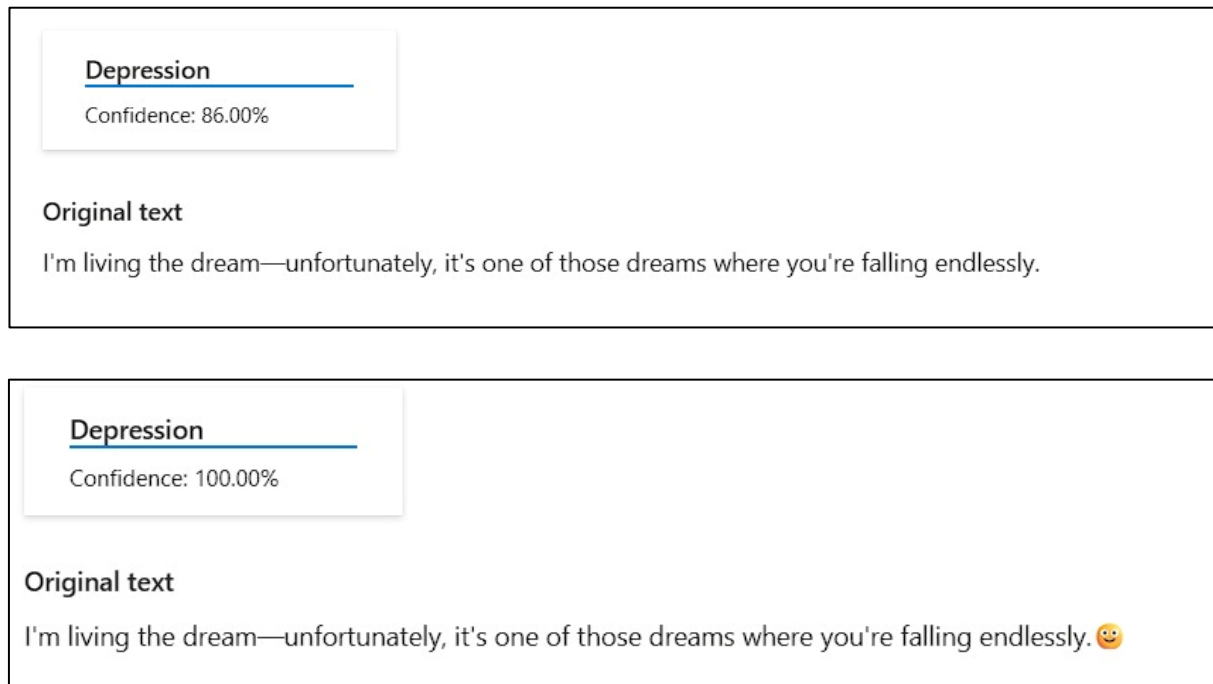


FIGURE.05: Feedback Generation Algorithm

Algorithm 1: Feedback Generation	
Input	: Predicted mental health condition $\hat{y}$, sentiment S
Output	: Personalized feedback message M and recommended resources R
<pre>1: Input predicted mental health condition $\hat{y}$ 2: Input sentiment S 3: Initialize message $M \leftarrow ""$ 4: Initialize resources $R \leftarrow \emptyset$ 5: if $\hat{y} = \text{Suicidal Ideation and Behavior}$ then 6: $M \leftarrow \text{"You are not alone. Please talk to someone or call a helpline."}$ 7: $R \leftarrow \text{crisis resources and helpline links}$ 8: else if $\hat{y} = \text{Depression}$ then 9: $M \leftarrow \text{"You are valuable. Consider seeking support for your well-being."}$ 10: $R \leftarrow \text{relaxation exercises and self-care resources}$ 11: else if $\hat{y} = \text{Anxiety}$ then 12: $M \leftarrow \text{"Try breathing exercises or mindfulness to calm down."}$ 13: $R \leftarrow \text{mindfulness techniques and anxiety-management resources}$ 14: else 15: $M \leftarrow \text{"You are doing great! Keep it up!"}$ 16: $R \leftarrow \text{positive reinforcement resources (motivation, encouragement)}$ 17: end if 18: Return M, R</pre>	

FIGURE.06: Admin Monitoring and Reporting Algorithm

Algorithm 2: Admin Monitoring and Reporting	
Input	: User records R stored in Firebase Realtime Database
Output	: Dashboard statistics D , User history H , Classification reports C
<pre> 1: Authenticate administrator credentials 2: Establish secure connection to Firebase Realtime Database 3: Retrieve all user records R 4: Initialize user history $H \leftarrow \emptyset$, sentiment count $S[3] \leftarrow \{0, 0, 0\}$ 5: Initialize category count $F[6] \leftarrow \{0, 0, 0, 0, 0, 0\}$ 6: for each record $r \in R$ do 7: Extract user_id, text, sentiment s, category c, timestamp t from r 8: Append (user_id, text, s, c, t) to user history H 9: Update sentiment count: $S[\text{index}(s)] \leftarrow S[\text{index}(s)] + 1$ 10: Update category count: $F[\text{index}(c)] \leftarrow F[\text{index}(c)] + 1$ 11: end for 12: Compute sentiment distribution: $SD[j] \leftarrow S[j] / \sum_{k=0}^2 S[k]$, for $j = 0, 1, 2$ 13: Compute category-wise frequencies: $CF[i] \leftarrow F[i] / \sum_{k=0}^5 F[k]$, for $i = 0, 1, \dots, 5$ 14: Generate trend analysis summaries based on H 15: Update dashboard visualizations D with SD, CF, and trends 16: Display reports C (user history, statistics, and trends) to administrator 17: Monitor database for new entries (real-time synchronization) 18: if new entry detected then 19: Add new record to R and update H 20: Recalculate SD and CF 21: Refresh dashboard visualizations D 22: Update reports C 23: end if 24: Repeat monitoring process continuously </pre>	
Notation: $S[3]$ – count of sentiments {Positive, Neutral, Negative}; $F[6]$ – count of categories	

TABLES:

TABLE.01: Class Type Performance (Classification Performance on the Test Dataset)

CLASS	PRECISION(%)	RECALL(%)	F1 SCORE
Anxiety	95.24	100.00	97.6
Depression	100.00	100.00	100
PTSD	100.00	95.00	97.4
Social Anxiety Disorder	90.48	100.00	95
Suicidal Ideation and Behavior	100.00	90.00	94.7

TABLE.02: Comparison with representative mental health classification studies reported

MODEL	REPORTED METRIC
<i>Bert</i> [35]	Precision = 73.34 ; Recall = 81.38; F1 Score = 76.69
<i>DistilBert</i> [35]	Precision = 72.95 ; Recall = 78.67; F1 Score = 75.43
<i>Persona-informed + Self-supervised</i> [35]	Precision = 65.59 ; Recall = 82.29; F1 Score = 79.18
<i>MultiResCNN</i> [36]	F1 Score =0.552
<i>EffectiveCAN (Multi-layer attention)</i> [36]	F1 Score =0.581
<i>EECAN (Multi-layer attention)</i> [36]	F1 Score = 0.585
PROPOSED MODEL	Precision = 96.97 % ; Recall = 96.97 ; F1 Score = 96.97%